



INTALG: Polynomials and Rational Expressions

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Learning Objectives

- Multiply polynomials using FOIL and special products
- Factor polynomials completely (GCF, trinomial, difference of squares)
- Simplify, multiply, and divide rational expressions
- Find the domain of a rational expression

Simplify each expression completely. Show all steps and circle your final answer.

Multiplying polynomials

1. Expand using FOIL: $(1x + 1)(3x + 2)$.

$$(1x + 1)(3x + 2)$$

Answer: _____

2. Expand using FOIL: $(3x + 4)(4x + 3)$.

$$(3x + 4)(4x + 3)$$

Answer: _____

3. Expand using FOIL: $(3x + 2)(3x + 7)$.

$$(3x + 2)(3x + 7)$$

Answer: _____

4. Expand using FOIL: $(2x + 7)(2x + 2)$.

$$(2x + 7)(2x + 2)$$

Answer: _____

5. Expand using FOIL: $(2x + 4)(3x + 4)$.

$$(2x + 4)(3x + 4)$$

Answer: _____

6. Expand using FOIL: $(1x + 3)(1x + 2)$.

$$(1x + 3)(1x + 2)$$

Answer: _____



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7. Expand using FOIL: $(3x + 3)(2x + 3)$.

$$(3x + 3)(2x + 3)$$

Answer: _____

8. Expand using FOIL: $(4x + 4)(3x + 3)$.

$$(4x + 4)(3x + 3)$$

Answer: _____

9. Expand using FOIL: $(1x + 3)(4x + 2)$.

$$(1x + 3)(4x + 2)$$

Answer: _____

10. Expand using FOIL: $(1x + 1)(2x + 1)$.

$$(1x + 1)(2x + 1)$$

Answer: _____

Rational expressions — domain

11. Find the excluded value(s) for $f(x) = (1x + -2) / (2x + 4)$.

$$f(x) = \frac{1x + -2}{2x + 4}$$

Answer: _____

12. Find the excluded value(s) for $f(x) = (1x + 0) / (1x + 8)$.

$$f(x) = \frac{1x + 0}{1x + 8}$$

Answer: _____

13. Find the excluded value(s) for $f(x) = (1x + -1) / (4x + -7)$.

$$f(x) = \frac{1x + -1}{4x + -7}$$

Answer: _____

14. Find the excluded value(s) for $f(x) = (3x + -6) / (5x + 5)$.

$$f(x) = \frac{3x + -6}{5x + 5}$$

Answer: _____



15. Find the excluded value(s) for $f(x) = (1x + -3) / (2x + -1)$.

$$f(x) = \frac{1x + -3}{2x + -1}$$

Answer: _____

16. Find the excluded value(s) for $f(x) = (5x + -3) / (2x + 10)$.

$$f(x) = \frac{5x + -3}{2x + 10}$$

Answer: _____

17. Find the excluded value(s) for $f(x) = (4x + -4) / (5x + -4)$.

$$f(x) = \frac{4x + -4}{5x + -4}$$

Answer: _____

18. Find the excluded value(s) for $f(x) = (4x + -5) / (5x + 2)$.

$$f(x) = \frac{4x + -5}{5x + 2}$$

Answer: _____

19. Find the excluded value(s) for $f(x) = (1x + -5) / (4x + 4)$.

$$f(x) = \frac{1x + -5}{4x + 4}$$

Answer: _____

20. Find the excluded value(s) for $f(x) = (1x + -2) / (4x + 4)$.

$$f(x) = \frac{1x + -2}{4x + 4}$$

Answer: _____

Simplifying rational expressions

21. Simplify: $(5x + 25) / (5x + 25)$. State any excluded values.

$$\frac{5x + 25}{5x + 25}$$

Answer: _____



22. Simplify: $(4x + 4) / (4x + 16)$. State any excluded values.

$$\frac{4x + 4}{4x + 16}$$

Answer: _____

23. Simplify: $(5x + 15) / (5x + 15)$. State any excluded values.

$$\frac{5x + 15}{5x + 15}$$

Answer: _____

24. Simplify: $(3x + 12) / (3x + 15)$. State any excluded values.

$$\frac{3x + 12}{3x + 15}$$

Answer: _____

25. Simplify: $(2x + 8) / (2x + 12)$. State any excluded values.

$$\frac{2x + 8}{2x + 12}$$

Answer: _____

26. Simplify: $(3x + 3) / (3x + 18)$. State any excluded values.

$$\frac{3x + 3}{3x + 18}$$

Answer: _____

27. Simplify: $(3x + 3) / (3x + 6)$. State any excluded values.

$$\frac{3x + 3}{3x + 6}$$

Answer: _____

28. Simplify: $(5x + 15) / (5x + 15)$. State any excluded values.

$$\frac{5x + 15}{5x + 15}$$

Answer: _____

29. Simplify: $(5x + 30) / (5x + 30)$. State any excluded values.

$$\frac{5x + 30}{5x + 30}$$

Answer: _____



30. Simplify: $(4x + 20) / (4x + 20)$. State any excluded values.

$$\frac{4x + 20}{4x + 20}$$

Answer: _____





Topics: Simplifying rational expressions, Rational expressions — domain, Multiplying polynomials. All answers verified by independent computation.

Solutions



Multiplying polynomials

1. Expand using FOIL: $(1x + 1)(3x + 2)$.

$$(1x + 1)(3x + 2)$$

$$\rightarrow F: 3x^2. O: 2x. I: 3x. L: 2.$$

$$\rightarrow = 3x^2 + 5x + 2.$$

Answer: $3x^2 + 5x + 2$

2. Expand using FOIL: $(3x + 4)(4x + 3)$.

$$(3x + 4)(4x + 3)$$

$$\rightarrow F: 12x^2. O: 9x. I: 16x. L: 12.$$

$$\rightarrow = 12x^2 + 25x + 12.$$

Answer: $12x^2 + 25x + 12$

3. Expand using FOIL: $(3x + 2)(3x + 7)$.

$$(3x + 2)(3x + 7)$$

$$\rightarrow F: 9x^2. O: 21x. I: 6x. L: 14.$$

$$\rightarrow = 9x^2 + 27x + 14.$$

Answer: $9x^2 + 27x + 14$

4. Expand using FOIL: $(2x + 7)(2x + 2)$.

$$(2x + 7)(2x + 2)$$

$$\rightarrow F: 4x^2. O: 4x. I: 14x. L: 14.$$

$$\rightarrow = 4x^2 + 18x + 14.$$

Answer: $4x^2 + 18x + 14$

5. Expand using FOIL: $(2x + 4)(3x + 4)$.

$$(2x + 4)(3x + 4)$$

$$\rightarrow F: 6x^2. O: 8x. I: 12x. L: 16.$$

$$\rightarrow = 6x^2 + 20x + 16.$$

Answer: $6x^2 + 20x + 16$

6. Expand using FOIL: $(1x + 3)(1x + 2)$.

$$(1x + 3)(1x + 2)$$

$$\rightarrow F: 1x^2. O: 2x. I: 3x. L: 6.$$

$$\rightarrow = 1x^2 + 5x + 6.$$

Answer: $1x^2 + 5x + 6$



7. Expand using FOIL: $(3x + 3)(2x + 3)$.

$$(3x + 3)(2x + 3)$$

→ F: $6x^2$. O: $9x$. I: $6x$. L: 9 .

→ = $6x^2 + 15x + 9$.

Answer: $6x^2 + 15x + 9$

8. Expand using FOIL: $(4x + 4)(3x + 3)$.

$$(4x + 4)(3x + 3)$$

→ F: $12x^2$. O: $12x$. I: $12x$. L: 12 .

→ = $12x^2 + 24x + 12$.

Answer: $12x^2 + 24x + 12$

9. Expand using FOIL: $(1x + 3)(4x + 2)$.

$$(1x + 3)(4x + 2)$$

→ F: $4x^2$. O: $2x$. I: $12x$. L: 6 .

→ = $4x^2 + 14x + 6$.

Answer: $4x^2 + 14x + 6$

10. Expand using FOIL: $(1x + 1)(2x + 1)$.

$$(1x + 1)(2x + 1)$$

→ F: $2x^2$. O: $1x$. I: $2x$. L: 1 .

→ = $2x^2 + 3x + 1$.

Answer: $2x^2 + 3x + 1$



Rational expressions — domain

11. Find the excluded value(s) for $f(x) = (1x + -2) / (2x + 4)$.

$$f(x) = \frac{1x + -2}{2x + 4}$$

→ Set $2x + 4 = 0$. Solve: $x = -2$.

Answer: $x \neq -2$

12. Find the excluded value(s) for $f(x) = (1x + 0) / (1x + 8)$.

$$f(x) = \frac{1x + 0}{1x + 8}$$

→ Set $1x + 8 = 0$. Solve: $x = -8$.

Answer: $x \neq -8$

13. Find the excluded value(s) for $f(x) = (1x + -1) / (4x + -7)$.

$$f(x) = \frac{1x + -1}{4x + -7}$$

→ Set $4x + -7 = 0$. Solve: $x = 7/4$.

Answer: $x \neq 7/4$

14. Find the excluded value(s) for $f(x) = (3x + -6) / (5x + 5)$.

$$f(x) = \frac{3x + -6}{5x + 5}$$

→ Set $5x + 5 = 0$. Solve: $x = -1$.

Answer: $x \neq -1$

15. Find the excluded value(s) for $f(x) = (1x + -3) / (2x + -1)$.

$$f(x) = \frac{1x + -3}{2x + -1}$$

→ Set $2x + -1 = 0$. Solve: $x = 1/2$.

Answer: $x \neq 1/2$

16. Find the excluded value(s) for $f(x) = (5x + -3) / (2x + 10)$.

$$f(x) = \frac{5x + -3}{2x + 10}$$

→ Set $2x + 10 = 0$. Solve: $x = -5$.

Answer: $x \neq -5$

17. Find the excluded value(s) for $f(x) = (4x + -4) / (5x + -4)$.

$$f(x) = \frac{4x + -4}{5x + -4}$$

→ Set $5x + -4 = 0$. Solve: $x = 4/5$.

Answer: $x \neq 4/5$



18. Find the excluded value(s) for $f(x) = (4x + -5) / (5x + 2)$.

$$f(x) = \frac{4x + -5}{5x + 2}$$

→ Set $5x + 2 = 0$. Solve: $x = -2/5$.

Answer: $x \neq -2/5$

19. Find the excluded value(s) for $f(x) = (1x + -5) / (4x + 4)$.

$$f(x) = \frac{1x + -5}{4x + 4}$$

→ Set $4x + 4 = 0$. Solve: $x = -1$.

Answer: $x \neq -1$

20. Find the excluded value(s) for $f(x) = (1x + -2) / (4x + 4)$.

$$f(x) = \frac{1x + -2}{4x + 4}$$

→ Set $4x + 4 = 0$. Solve: $x = -1$.

Answer: $x \neq -1$



Simplifying rational expressions

21. Simplify: $(5x + 25) / (5x + 25)$. State any excluded values.

$$\frac{5x + 25}{5x + 25}$$

→ Factor: $5(x + 5) / 5(x + 5)$.

→ Cancel 5: $(x + 5)/(x + 5)$, $x \neq -5$.

Answer: $\frac{x + 5}{x + 5}$, $x \neq -5$

22. Simplify: $(4x + 4) / (4x + 16)$. State any excluded values.

$$\frac{4x + 4}{4x + 16}$$

→ Factor: $4(x + 1) / 4(x + 4)$.

→ Cancel 4: $(x + 1)/(x + 4)$, $x \neq -4$.

Answer: $\frac{x + 1}{x + 4}$, $x \neq -4$

23. Simplify: $(5x + 15) / (5x + 15)$. State any excluded values.

$$\frac{5x + 15}{5x + 15}$$

→ Factor: $5(x + 3) / 5(x + 3)$.

→ Cancel 5: $(x + 3)/(x + 3)$, $x \neq -3$.

Answer: $\frac{x + 3}{x + 3}$, $x \neq -3$

24. Simplify: $(3x + 12) / (3x + 15)$. State any excluded values.

$$\frac{3x + 12}{3x + 15}$$

→ Factor: $3(x + 4) / 3(x + 5)$.

→ Cancel 3: $(x + 4)/(x + 5)$, $x \neq -5$.

Answer: $\frac{x + 4}{x + 5}$, $x \neq -5$

25. Simplify: $(2x + 8) / (2x + 12)$. State any excluded values.

$$\frac{2x + 8}{2x + 12}$$

→ Factor: $2(x + 4) / 2(x + 6)$.

→ Cancel 2: $(x + 4)/(x + 6)$, $x \neq -6$.

Answer: $\frac{x + 4}{x + 6}$, $x \neq -6$



26. Simplify: $(3x + 3) / (3x + 18)$. State any excluded values.

$$\frac{3x + 3}{3x + 18}$$

→ Factor: $3(x + 1) / 3(x + 6)$.

→ Cancel 3: $(x + 1)/(x + 6)$, $x \neq -6$.

Answer: $\frac{x + 1}{x + 6}$, $x \neq -6$

27. Simplify: $(3x + 3) / (3x + 6)$. State any excluded values.

$$\frac{3x + 3}{3x + 6}$$

→ Factor: $3(x + 1) / 3(x + 2)$.

→ Cancel 3: $(x + 1)/(x + 2)$, $x \neq -2$.

Answer: $\frac{x + 1}{x + 2}$, $x \neq -2$

28. Simplify: $(5x + 15) / (5x + 15)$. State any excluded values.

$$\frac{5x + 15}{5x + 15}$$

→ Factor: $5(x + 3) / 5(x + 3)$.

→ Cancel 5: $(x + 3)/(x + 3)$, $x \neq -3$.

Answer: $\frac{x + 3}{x + 3}$, $x \neq -3$

29. Simplify: $(5x + 30) / (5x + 30)$. State any excluded values.

$$\frac{5x + 30}{5x + 30}$$

→ Factor: $5(x + 6) / 5(x + 6)$.

→ Cancel 5: $(x + 6)/(x + 6)$, $x \neq -6$.

Answer: $\frac{x + 6}{x + 6}$, $x \neq -6$

30. Simplify: $(4x + 20) / (4x + 20)$. State any excluded values.

$$\frac{4x + 20}{4x + 20}$$

→ Factor: $4(x + 5) / 4(x + 5)$.

→ Cancel 4: $(x + 5)/(x + 5)$, $x \neq -5$.

Answer: $\frac{x + 5}{x + 5}$, $x \neq -5$

